SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I T. Pradeep (192411294) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Pradeep.

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%	2	2	2
Concept Mapping Structure	25%	2	2	2
Linking & Relationships	20%	2	2	2
Application & Analysis	20%	1	1	1
Explanation & Clarity	15%	1	1	1

Total Marks 24 /30

Signature of the Faculty

CONCEPT MAPPING QUESTIONS

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

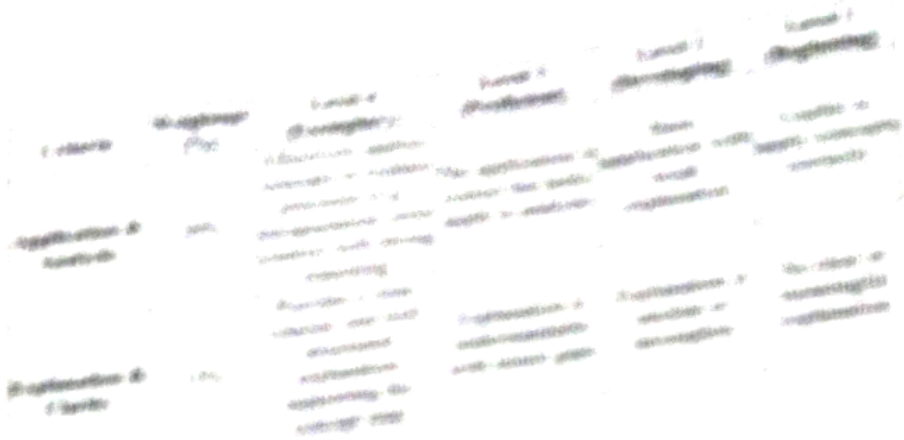
6. A network issue is represented in the concept map:

Physical ✔ → Data Link ✔ → Network ✘ → Transport ✘ → Application ✘

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships



How Data link → MAC address → local Delivery → Network → IP → Global

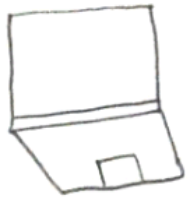
① Data link



works at the Data link layer

The data is first Packed into a small frames at the data link layer. This frames helps the network under-stand where local link.

② MAC address



MAC : AA : BB : CC : DD : FF

The frames uses a MAC address to identify the exact device on the local network.

This marks sure that the data reaches the right device.

③ local Delivery Network



Find the next network

The data is delivered within the local network usin MAC address

The router then checks the IP address where send it next.

⑤ Global Delivery

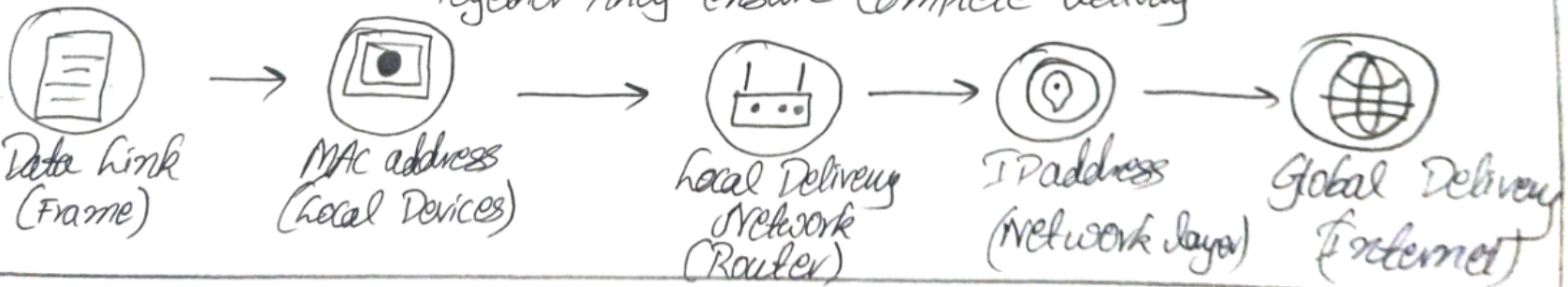


Reaches the final destination

Finally, the data travels across multiple networks and routers until it reaches the destination's IP address

This mean the data delivery globally.

Together, they ensure complete delivery



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Web Browsing using OSI Layers

The OSI Layers in web Browsing

⑦ Application layer :-
Handles web services browser requests, web content

⑥ Presentation layer :-
Formats data (HTML, CSS, JS)
encryption (TLS/SSL)

⑤ Session layer :-
Maintains the connection
(eg. - session mangement, cookies)

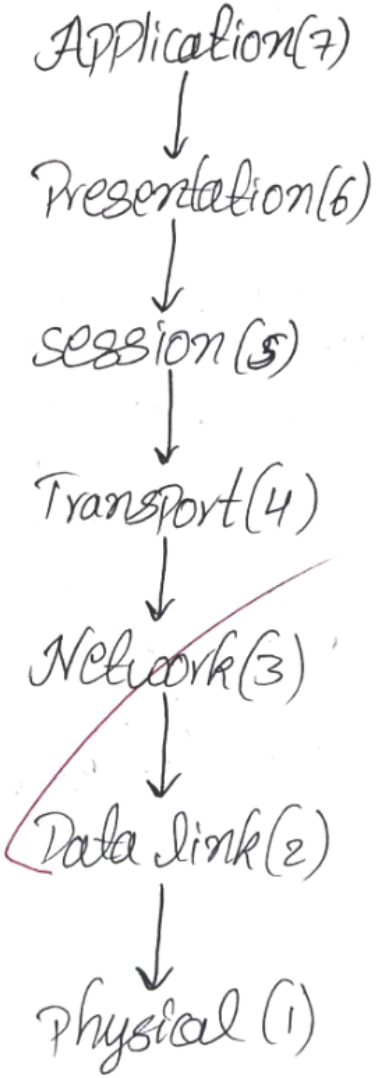
④ Transport layer :-
Provides end to end delivery

③ Network layer :-
* Routes data

② Data link layer :-
Handles local delivery

① Physical layer :-
Send raw bits over the medium

How it works



what if one layer fails?

7) Application layer → Browser can't find the website load page

6) Presentation layer → Data can't be read or verified

5) session layer → Connection get dropped no data

4) Transport layer → Data is not delivered reliability

3) Network layer → Packets can't reach the destin

2) Data link layer → Devices can't connect to the network

1) Physical layer → No signal.

Network Issue Analysis and Troubleshooting

The Given Concept Map

✓ = working X = Failing

Physical layer (✓)
Link, cable, signal
hardwork work file

→

Data link layer (✓)
Frames are created and delived to the next hop

→

Network layer (X)
Cannot route Packets blw different networks

→

Transport layer (X)
Cannot establish end-to-end connection (eg:- TCP/UDP)

→

Application layer (X)
web services and application cannot be reached

How the concept Map Helps in Trouble shooting

① Start from the bottom
Since Physical and Data link are working, we can rule out cable, signal and local frame issue

② Focus on the Network layer
Checks IP address, sub-net mask, default gate-way and routing table verify the router or L3 device is working

③ Check the Transport layer
After fixing the network layer, verify if the correct Port are open and if there firewall.

④ Fix the Application layer
Once transport works, check the web server DNS, and application.